Class 12 Mathematics

Chapter 7: Integrals

Practice Set A: step-wise solutions

20 questions **54** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 MCQ PYQ-style — 1 mark

$\int e^x \sec x\,(1+\tan x)\,dx$ is equal to:

(A) $e^x \cos x + C$ **(B)** $e^x \sec x + C$ **(C)** $e^x \sin x + C$ **(D)** $e^x \tan x + C$

Formula / idea: $\int e^x\,[f(x)+f'(x)]\,dx = e^x f(x) + C$

1 Open the bracket
$e^x \sec x\,(1+\tan x) = e^x\,(\sec x + \sec x \tan x)$

2 Take $f(x) = \sec x$
$f'(x) = \sec x \tan x$

3 So the integral is
$\int e^x(\sec x + \sec x \tan x)\,dx = e^x \sec x + C$ ✓ 1

Answer **(B)** $e^x \sec x + C$

Watch out: Picking (D): $f(x) = \tan x$ needs $f'(x) = \sec^2 x$, which is not in the bracket.

2 MCQ CBSE SQP 2025-26 — 1 mark

If $\int x^3 \sin^4(x^4)\cos(x^4)\,dx = a\sin^5(x^4) + C$, then a is equal to:

(A) $-\frac{1}{10}$ **(B)** $\frac{1}{20}$ **(C)** $\frac{1}{4}$ **(D)** $\frac{1}{5}$

Formula / idea: Put the inside function $= t$; its derivative must be present (up to a constant).

1 Put $\sin(x^4) = t$
$\cos(x^4)\cdot 4x^3\,dx = dt \;\Rightarrow\; x^3\cos(x^4)\,dx = \frac{dt}{4}$

2 Substitute

$$\int t^4 \cdot \frac{dt}{4} = \frac{1}{4} \cdot \frac{t^5}{5} + C = \frac{t^5}{20} + C$$

3 Put back $t = \sin(x^4)$ and compare

$$\frac{1}{20}\sin^5(x^4) + C \;\Rightarrow\; a = \frac{1}{20}$$

✓ 1

Answer **(B)** $\frac{1}{20}$

Watch out: Forgetting the $\frac{1}{4}$ from $d(x^4) = 4x^3\,dx$ and choosing (D) $\frac{1}{5}$.

3 MCQ PYQ-style 1 mark

The value of $\int_0^{\pi/2} x \sin x \, dx$ is:

(A) 0 **(B)** 1 **(C)** $\frac{\pi}{2}$ **(D)** $\frac{\pi}{2} - 1$

Formula / idea: By parts: $\int u\,v\,dx = u\int v\,dx - \int \left(u' \int v\,dx\right) dx$

1 Integrate by parts, taking $u = x$, $v = \sin x$

$$\int x \sin x \, dx = -x\cos x + \int \cos x \, dx = -x\cos x + \sin x$$

2 Put the limits

$$\left[-x\cos x + \sin x\right]_0^{\pi/2} = \left(-\frac{\pi}{2} \cdot 0 + 1\right) - (0 + 0)$$

3 Simplify

$$= 1$$

✓ 1

Answer **(B)** 1

Watch out: Writing $\int \sin x\,dx = \cos x$; the correct result is $-\cos x$.

4 MCQ CBSE SQP 2024-25 1 mark

$\int \dfrac{dx}{x^3\,(1+x^4)^{1/2}}$ equals:

(A) $-\dfrac{1}{2x^2}\sqrt{1+x^4}+C$ **(B)** $\dfrac{1}{2x}\sqrt{1+x^4}+C$

(C) $-\dfrac{1}{4x}\sqrt{1+x^4}+C$ **(D)** $\dfrac{1}{4x^2}\sqrt{1+x^4}+C$

Formula / idea: Take the highest power of x out of the root, then substitute.

1 Take x^4 out of the root; $\sqrt{x^4}=x^2$

$$\int \frac{dx}{x^3\cdot x^2\sqrt{1+x^{-4}}}=\int \frac{x^{-5}\,dx}{\sqrt{1+x^{-4}}}$$

2 Put $1+x^{-4}=t$

$$-4x^{-5}\,dx=dt$$

3 Substitute

$$-\frac{1}{4}\int t^{-1/2}\,dt=-\frac{1}{2}\sqrt{t}+C$$

4 Put back $t=\dfrac{1+x^4}{x^4}$ ✓ 1

$$-\frac{1}{2}\cdot\frac{\sqrt{1+x^4}}{x^2}+C=-\frac{1}{2x^2}\sqrt{1+x^4}+C$$

Answer **(A)** $-\dfrac{1}{2x^2}\sqrt{1+x^4}+C$

Watch out: Dropping the minus sign in $dt=-4x^{-5}\,dx$, or the $\frac{1}{4}$ with it; the answer is $-\dfrac{1}{2x^2}\sqrt{1+x^4}$.

5 MCQ CBSE SQP 2025-26 1 mark

If $f(a+b-x)=f(x)$, then $\int_a^b x\,f(x)\,dx$ is equal to:

(A) $\dfrac{a+b}{2}\int_a^b f(b-x)\,dx$ **(B)** $\dfrac{a+b}{2}\int_a^b f(a-x)\,dx$

(C) $\dfrac{b-a}{2}\int_a^b f(x)\,dx$ **(D)** $\dfrac{a+b}{2}\int_a^b f(x)\,dx$

Formula / idea: $\displaystyle\int_a^b f(x)\,dx=\int_a^b f(a+b-x)\,dx$

1 Let

$$I=\int_a^b x\,f(x)\,dx \quad \ldots(1)$$

2 Replace x by $a+b-x$ and use $f(a+b-x)=f(x)$

$$I=\int_a^b (a+b-x)\,f(x)\,dx \quad \dots(2)$$

3 Add (1) and (2) ✓ 1

$$2I=(a+b)\int_a^b f(x)\,dx \;\Rightarrow\; I=\frac{a+b}{2}\int_a^b f(x)\,dx$$

Answer **(D)** $\dfrac{a+b}{2}\displaystyle\int_a^b f(x)\,dx$

Watch out: Choosing (C): the factor is $a+b$ (from adding x and $a+b-x$), not $b-a$.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): $\displaystyle\int_{-1}^{1}\log\left(\frac{2-x}{2+x}\right)dx=0$. **Reason (R):** If f is an even function, then $\displaystyle\int_{-a}^{a}f(x)\,dx=2\int_0^a f(x)\,dx$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) **Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: If $f(-x)=-f(x)$, then $\displaystyle\int_{-a}^{a}f(x)\,dx=0$

1 Check A: let $f(x)=\log\dfrac{2-x}{2+x}$ and replace x by $-x$

$$f(-x)=\log\frac{2+x}{2-x}=-\log\frac{2-x}{2+x}=-f(x)$$

2 So f is odd; its integral over $[-1,1]$ is 0. A is true.

3 R is the standard property of even functions. R is true.

4 A uses the odd-function property, not R. So R does not explain A.

5 Hence the correct option is (B). ✓ 1

Answer **(B) Both true; R is not the correct explanation of A**

Watch out: Marking (A) because both statements are true, without checking that R is about even functions.

7 **Short answer** PYQ-style 2 marks

Find $\int \frac{\sin x - \cos x}{\sqrt{1+\sin 2x}}\,dx,\ 0 < x < \frac{\pi}{2}$.

Formula / idea: $1 + \sin 2x = (\sin x + \cos x)^2$

1 Write $1 = \sin^2 x + \cos^2 x$ and $\sin 2x = 2\sin x \cos x$
$1 + \sin 2x = (\sin x + \cos x)^2$ ✓ ½

2 For $0 < x < \frac{\pi}{2}$, $\sin x + \cos x > 0$
$I = \int \frac{\sin x - \cos x}{\sin x + \cos x}\,dx$ ✓ ½

3 Put $\sin x + \cos x = t$
$(\cos x - \sin x)\,dx = dt \ \Rightarrow\ (\sin x - \cos x)\,dx = -dt$ ✓ ½

4 Integrate and put back t
$I = -\int \frac{dt}{t} = -\log|t| + C = -\log|\sin x + \cos x| + C$ ✓ ½

Answer $-\log|\sin x + \cos x| + C$

Watch out: Losing the minus sign: the derivative of $\sin x + \cos x$ is $\cos x - \sin x$, the negative of the numerator.

Level 2

19 marks

8 **Short answer** PYQ-style 2 marks

Evaluate: $\int_0^{\pi/4} \tan^2 x\,dx$.

Formula / idea: $\tan^2 x = \sec^2 x - 1$

1 Use $\tan^2 x = \sec^2 x - 1$
$I = \int_0^{\pi/4} (\sec^2 x - 1)\,dx$ ✓ ½

2 Integrate
$I = \left[\tan x - x\right]_0^{\pi/4}$ ✓ ½

3 Put the limits

$$I = \left(\tan\frac{\pi}{4} - \frac{\pi}{4}\right) - (0 - 0) = 1 - \frac{\pi}{4}$$

✓ 1

Answer $1 - \frac{\pi}{4}$

Watch out: Writing $\int \tan^2 x\, dx = \frac{\tan^3 x}{3}$; there is no $\sec^2 x$ factor for that.

9 **Short answer** CBSE SQP 2023-24 2 marks

Find: $\int \sqrt{\frac{x}{1-x^3}}\, dx;\ x \in (0,1).$

Formula / idea: $\int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1} t + C$

1 Write $x^3 = \left(x^{3/2}\right)^2$

$$I = \int \frac{\sqrt{x}}{\sqrt{1-\left(x^{3/2}\right)^2}}\, dx$$

✓ ½

2 Put $x^{3/2} = t$

$$\frac{3}{2}x^{1/2}\, dx = dt \ \Rightarrow\ \sqrt{x}\, dx = \frac{2}{3}\, dt$$

✓ ½

3 Substitute

$$I = \frac{2}{3}\int \frac{dt}{\sqrt{1-t^2}} = \frac{2}{3}\sin^{-1} t + C$$

✓ ½

4 Put back $t = x^{3/2}$

$$I = \frac{2}{3}\sin^{-1}\left(x^{3/2}\right) + C$$

✓ ½

Answer $\frac{2}{3}\sin^{-1}\left(x^{3/2}\right) + C$

Watch out: Putting $x^3 = t$; then $\sqrt{x}$ in the numerator does not match $3x^2\, dx$.

10 **Short answer** CBSE SQP 2023-24 3 marks

Find: $\int \frac{2x^2+3}{x^2\left(x^2+9\right)}\, dx;\ x \neq 0.$

Formula / idea: Only x^2 appears, so put $x^2 = y$ just to find the partial fractions.

1 Put $x^2 = y$ (for partial fractions only)

$$\frac{2y+3}{y(y+9)} = \frac{A}{y} + \frac{B}{y+9}$$

✓ ½

2 Multiply by $y(y+9)$
$2y+3 = A(y+9)+By$ ✓ ½

3 Put $y=0$, then $y=-9$
$A = \frac{3}{9} = \frac{1}{3}, \qquad B = \frac{-15}{-9} = \frac{5}{3}$ ✓ 1

4 Put back $y = x^2$ and integrate
$I = \frac{1}{3}\int x^{-2}\,dx + \frac{5}{3}\int \frac{dx}{x^2+3^2}$ ✓ ½

5 Use $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$
$I = -\frac{1}{3x} + \frac{5}{9}\tan^{-1}\frac{x}{3} + C$ ✓ ½

Answer $-\frac{1}{3x} + \frac{5}{9}\tan^{-1}\frac{x}{3} + C$

Watch out: Integrating $\frac{1}{x^2+9}$ as $\tan^{-1}\frac{x}{3}$ and dropping the factor $\frac{1}{3}$.

11 **Short answer** PYQ-style 3 marks

Evaluate: $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$.

Formula / idea: $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$; here $a+b = \frac{\pi}{2}$.

1 Write $\tan x = \frac{\sin x}{\cos x}$
$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx \quad \ldots(1)$ ✓ ½

2 Replace x by $\frac{\pi}{6}+\frac{\pi}{3}-x = \frac{\pi}{2}-x$
$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx \quad \ldots(2)$ ✓ 1

3 Add (1) and (2)
$2I = \int_{\pi/6}^{\pi/3} 1\,dx$ ✓ ½

4 Integrate
$2I = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$ ✓ ½

5 Hence
$I = \dfrac{\pi}{12}$ ✓ ½

Answer $\dfrac{\pi}{12}$

Watch out: Replacing x by $\dfrac{\pi}{3} - x$; with limits a to b it must be $a + b - x$.

12 **Short answer** CBSE SQP 2024-25 3 marks

Evaluate: $\displaystyle\int \left\{ \frac{1}{\log x} - \frac{1}{(\log x)^2} \right\} dx;\ x > 1.$

Formula / idea: $\displaystyle\int e^t \left[f(t) + f'(t) \right] dt = e^t f(t) + C$

1 Put $\log x = t$
$x = e^t, \qquad dx = e^t\, dt$ ✓ 1

2 Substitute
$I = \displaystyle\int e^t \left(\frac{1}{t} - \frac{1}{t^2} \right) dt$ ✓ ½

3 Take $f(t) = \dfrac{1}{t}$
$f'(t) = -\dfrac{1}{t^2}$ ✓ ½

4 Use the $e^t[f + f']$ result
$I = e^t \cdot \dfrac{1}{t} + C$ ✓ ½

5 Put back $e^t = x$, $t = \log x$
$I = \dfrac{x}{\log x} + C$ ✓ ½

Answer $\dfrac{x}{\log x} + C$

Watch out: Forgetting $dx = e^t\, dt$, which is what creates the e^t factor.

13 **Short answer** CBSE SQP 2025-26 3 marks

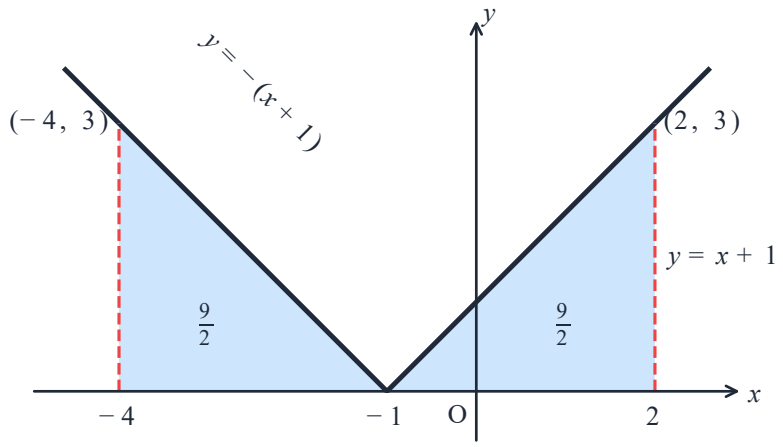
Sketch the graph of $y = |x + 1|$. Evaluate $\int_{-4}^{2} |x + 1|\, dx$. What does the value of this integral represent on the graph?

Formula / idea: Split a modulus integral at the point where the expression inside is 0.

1 Remove the modulus ($x + 1 = 0$ at $x = -1$) ✓ ½

$$|x+1| = \begin{cases} -(x+1), & x < -1 \\ x+1, & x \geq -1 \end{cases}$$

2 Sketch: a V-shape with vertex at $(-1, 0)$ ✓ ½



3 Split the integral at $x = -1$ ✓ ½

$$I = \int_{-4}^{-1} -(x+1)\, dx + \int_{-1}^{2} (x+1)\, dx$$

4 First part ✓ ½

$$-\left[\frac{(x+1)^2}{2}\right]_{-4}^{-1} = -\left(0 - \frac{9}{2}\right) = \frac{9}{2}$$

5 Second part ✓ ½

$$\left[\frac{(x+1)^2}{2}\right]_{-1}^{2} = \frac{9}{2} - 0 = \frac{9}{2}$$

6 Add; the value is the shaded area in the graph ✓ ½

$$I = \frac{9}{2} + \frac{9}{2} = 9 \text{ sq. units}$$

Answer **9; it is the area bounded by $y = |x + 1|$, the x-axis and the lines $x = -4$, $x = 2$.**

Watch out: Integrating $(x + 1)$ straight from -4 to 2, which gives 3 because the left part counts as negative.

14 **Short answer** CBSE 2025 3 marks

Evaluate: $\displaystyle\int_0^{\pi/4} \frac{dx}{\cos^3 x\sqrt{2\sin 2x}}$.

Formula / idea: Change everything to $\tan x$ and $\sec^2 x$, then put $\tan x = t$.

1 Use $\sin 2x = 2\sin x\cos x = 2\tan x\cos^2 x$
$\sqrt{2\sin 2x} = \sqrt{4\tan x\cos^2 x} = 2\cos x\sqrt{\tan x}$ ✓ ½

2 So the integrand becomes
$\displaystyle\frac{1}{2\cos^4 x\sqrt{\tan x}} = \frac{\sec^2 x\,(1+\tan^2 x)}{2\sqrt{\tan x}}$ ✓ ½

3 Put $\tan x = t$, $\sec^2 x\,dx = dt$
$x = 0 \Rightarrow t = 0, \qquad x = \dfrac{\pi}{4} \Rightarrow t = 1$ ✓ ½

4 Change the integral
$\displaystyle I = \frac{1}{2}\int_0^1 \left(t^{-1/2} + t^{3/2}\right) dt$ ✓ ½

5 Integrate
$\displaystyle I = \frac{1}{2}\left[2t^{1/2} + \frac{2}{5}t^{5/2}\right]_0^1$ ✓ ½

6 Put the limits
$\displaystyle I = \frac{1}{2}\left(2 + \frac{2}{5}\right) = \frac{6}{5}$ ✓ ½

Answer $\dfrac{6}{5}$

Watch out: Writing $\sec^4 x = \sec^2 x$; one $\sec^2 x$ must become $1 + \tan^2 x$.

Level 3

27 marks

15 **Long answer** PYQ-style 5 marks

Evaluate: $\displaystyle\int_0^{\pi} \frac{x\tan x}{\sec x + \tan x}\,dx$.

Formula / idea: $\displaystyle\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ removes the factor x.

1 Let
$\displaystyle I = \int_0^{\pi} \frac{x\tan x}{\sec x + \tan x}\,dx \quad \ldots(1)$

2 Replace x by $\pi - x$; tan and sec both change sign ✓ 1

$$I = \int_0^{\pi} \frac{(\pi - x)(-\tan x)}{-\sec x - \tan x}\,dx = \int_0^{\pi} \frac{(\pi - x)\tan x}{\sec x + \tan x}\,dx \quad \dots(2)$$

3 Add (1) and (2) ✓ ½

$$2I = \pi \int_0^{\pi} \frac{\tan x}{\sec x + \tan x}\,dx$$

4 Multiply top and bottom by $\cos x$ ✓ ½

$$\frac{\tan x}{\sec x + \tan x} = \frac{\sin x}{1 + \sin x} = 1 - \frac{1}{1 + \sin x}$$

5 Multiply by $\dfrac{1 - \sin x}{1 - \sin x}$ ✓ ½

$$\frac{1}{1 + \sin x} = \frac{1 - \sin x}{\cos^2 x} = \sec^2 x - \sec x \tan x$$

6 So ✓ ½

$$2I = \pi \int_0^{\pi} \left(1 - \sec^2 x + \sec x \tan x\right) dx$$

7 Integrate ✓ 1

$$2I = \pi \left[x - \tan x + \sec x\right]_0^{\pi}$$

$\sec x - \tan x = \dfrac{\cos x}{1 + \sin x}$ has no break at $\dfrac{\pi}{2}$

8 Put the limits ($\sec \pi = -1$, $\sec 0 = 1$) ✓ ½

$$2I = \pi\left[(\pi - 0 - 1) - (0 - 0 + 1)\right] = \pi(\pi - 2)$$

9 Hence ✓ ½

$$I = \frac{\pi(\pi - 2)}{2}$$

Answer $\dfrac{\pi(\pi - 2)}{2}$

Watch out: Taking $\sec(\pi - x) = \sec x$; it is $-\sec x$, like $\cos(\pi - x) = -\cos x$.

16 **Long answer** CBSE SQP 2025-26 5 marks

Find: $\displaystyle\int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta}\,d\theta.$

Formula / idea: Repeated factor $(t - a)^2$ needs two fractions: $\dfrac{A}{t - a} + \dfrac{B}{(t - a)^2}$.

1 Put $\sin\theta = t$ ✓ 1

$$\cos\theta\,d\theta = dt$$

2 Use $\cos^2\theta = 1 - t^2$ in the denominator ✓ ½

$$5 - (1 - t^2) - 4t = t^2 - 4t + 4 = (t - 2)^2$$

3 So
$$I = \int \frac{3t-2}{(t-2)^2}\,dt$$
✓ ½

4 Write as partial fractions
$$\frac{3t-2}{(t-2)^2} = \frac{A}{t-2} + \frac{B}{(t-2)^2} \Rightarrow 3t-2 = A(t-2)+B$$
✓ ½

5 Compare coefficient of t; then put $t = 2$
$$A = 3, \qquad B = 4$$
✓ 1

6 Integrate each fraction
$$I = 3\int \frac{dt}{t-2} + 4\int (t-2)^{-2}\,dt = 3\log|t-2| - \frac{4}{t-2} + C$$
✓ 1

7 Put back $t = \sin\theta$; note $|\sin\theta - 2| = 2 - \sin\theta$
$$I = 3\log(2-\sin\theta) + \frac{4}{2-\sin\theta} + C$$
✓ ½

Answer $3\log(2-\sin\theta) + \dfrac{4}{2-\sin\theta} + C$

Watch out: Writing $\int \frac{dt}{(t-2)^2} = \log(t-2)^2$; it is $-\frac{1}{t-2}$.

17 **Long answer** CBSE SQP 2025-26 5 marks

Evaluate: $\displaystyle\int_0^1 \frac{\log(1+x)}{1+x^2}\,dx.$

Formula / idea: Put $x = \tan\theta$, then use $\displaystyle\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx.$

1 Put $x = \tan\theta$, $dx = \sec^2\theta\,d\theta$
$$x = 0 \Rightarrow \theta = 0, \qquad x = 1 \Rightarrow \theta = \frac{\pi}{4}$$
✓ 1

2 Use $1 + \tan^2\theta = \sec^2\theta$
$$I = \int_0^{\pi/4} \frac{\log(1+\tan\theta)}{\sec^2\theta}\sec^2\theta\,d\theta = \int_0^{\pi/4} \log(1+\tan\theta)\,d\theta \quad \ldots(1)$$
✓ ½

3 Replace θ by $\frac{\pi}{4} - \theta$
$$1 + \tan\left(\frac{\pi}{4} - \theta\right) = 1 + \frac{1-\tan\theta}{1+\tan\theta} = \frac{2}{1+\tan\theta}$$
✓ 1

4 So
$$I = \int_0^{\pi/4} \log\frac{2}{1+\tan\theta}\,d\theta = \int_0^{\pi/4} [\log 2 - \log(1+\tan\theta)]\,d\theta$$
✓ ½

5 Use (1) for the second integral
$$I = \frac{\pi}{4}\log 2 - I$$
✓ 1

6 Hence ✓ 1

$$2I = \frac{\pi}{4}\log 2 \;\Rightarrow\; I = \frac{\pi}{8}\log 2$$

Answer $\frac{\pi}{8}\log 2$

Watch out: Not changing the limits: after $x = \tan\theta$ the upper limit is $\frac{\pi}{4}$, not 1.

18 **Short answer** PYQ-style 3 marks

Find: $\int \frac{x+3}{\sqrt{5-4x-x^2}}\,dx.$

Formula / idea: Numerator $= A\cdot\frac{d}{dx}(\text{quadratic}) + B$; then $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}$

1 Write $x+3 = A\frac{d}{dx}(5-4x-x^2) + B$

$x+3 = A(-4-2x) + B$

2 Compare coefficients: $-2A = 1$, $-4A + B = 3$ ✓ ½

$A = -\frac{1}{2}, \qquad B = 1$

3 Split the integral ✓ ½

$$I = -\frac{1}{2}\int \frac{-4-2x}{\sqrt{5-4x-x^2}}\,dx + \int \frac{dx}{\sqrt{5-4x-x^2}} = -\frac{1}{2}I_1 + I_2$$

4 In I_1 put $5-4x-x^2 = t$, so $(-4-2x)\,dx = dt$ ✓ ½

$$I_1 = \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} = 2\sqrt{5-4x-x^2}$$

5 In I_2 complete the square ✓ ½

$5-4x-x^2 = 9-(x+2)^2 = 3^2-(x+2)^2$

6 Use $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}$ ✓ ½

$I_2 = \sin^{-1}\left(\frac{x+2}{3}\right)$

7 Hence ✓ ½

$$I = -\sqrt{5-4x-x^2} + \sin^{-1}\left(\frac{x+2}{3}\right) + C$$

Answer $-\sqrt{5-4x-x^2} + \sin^{-1}\left(\frac{x+2}{3}\right) + C$

Watch out: Completing the square wrongly as $9+(x+2)^2$; take out the minus sign first: $-(x^2+4x-5)$.

19 **Long answer** PYQ-style 5 marks

Evaluate: $\displaystyle\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16\sin 2x}\,dx.$

Formula / idea: Put $\sin x - \cos x = t$; then $\sin 2x = 1 - t^2$.

1	Put $\sin x - \cos x = t$ $(\cos x + \sin x)\,dx = dt$	✓ 1
2	Square $\sin x - \cos x = t$ $1 - \sin 2x = t^2 \;\Rightarrow\; \sin 2x = 1 - t^2$	✓ 1
3	Change the limits $x = 0 \Rightarrow t = -1, \qquad x = \dfrac{\pi}{4} \Rightarrow t = 0$	✓ ½
4	Simplify the denominator $9 + 16(1 - t^2) = 25 - 16t^2$	✓ ½
5	So $I = \displaystyle\int_{-1}^{0} \frac{dt}{25 - 16t^2} = \frac{1}{16}\int_{-1}^{0} \frac{dt}{\left(\frac{5}{4}\right)^2 - t^2}$	✓ ½
6	Use $\displaystyle\int \frac{dt}{a^2 - t^2} = \frac{1}{2a}\log\left\lvert\frac{a+t}{a-t}\right\rvert$ with $a = \dfrac{5}{4}$ $I = \dfrac{1}{16}\cdot\dfrac{2}{5}\left[\log\left\lvert\dfrac{5+4t}{5-4t}\right\rvert\right]_{-1}^{0} = \dfrac{1}{40}\left[\log\left\lvert\dfrac{5+4t}{5-4t}\right\rvert\right]_{-1}^{0}$	✓ 1
7	Put the limits $I = \dfrac{1}{40}\left[\log 1 - \log\dfrac{1}{9}\right] = \dfrac{1}{40}\log 9$	✓ ½

Answer $\dfrac{1}{40}\log 9$

Watch out: Putting $\sin x + \cos x = t$; its derivative is $\cos x - \sin x$, which does not match the numerator.

20 **Case study** Board-pattern 4 marks

Case Study: While revising definite integrals, a student notes the property $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$. When the lower limit is 0, it becomes $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. Use these properties to answer the following.

(i) Evaluate $\displaystyle\int_2^8 \frac{\sqrt{10-x}}{\sqrt{x}+\sqrt{10-x}}\,dx$. 1 mark

(ii) Evaluate $\displaystyle\int_0^{\pi/2} \log(\tan x)\,dx$. 1 mark

(iii) Evaluate $\displaystyle\int_0^{\pi/2} (2\log\sin x - \log\sin 2x)\,dx$. 2 marks

Formula / idea: $\displaystyle\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$; on $\left[0, \frac{\pi}{2}\right]$, sin and cos swap.

(i) 1 mark

1 Let

$$I = \int_2^8 \frac{\sqrt{10-x}}{\sqrt{x}+\sqrt{10-x}}\,dx \quad \ldots(1)$$

2 Replace x by $2+8-x = 10-x$ ✓ ½

$$I = \int_2^8 \frac{\sqrt{x}}{\sqrt{10-x}+\sqrt{x}}\,dx \quad \ldots(2)$$

3 Add (1) and (2) ✓ ½

$$2I = \int_2^8 1\,dx = 6 \;\Rightarrow\; I = 3$$

Answer (i) 3

(ii) 1 mark

1 Let

$$I = \int_0^{\pi/2} \log(\tan x)\,dx \quad \ldots(1)$$

2 Replace x by $\frac{\pi}{2} - x$ ✓ ½

$$I = \int_0^{\pi/2} \log(\cot x)\,dx = -\int_0^{\pi/2} \log(\tan x)\,dx = -I$$

3 So ✓ ½

$$2I = 0 \;\Rightarrow\; I = 0$$

Answer (ii) 0

(iii) 2 marks

1	Use $\sin 2x = 2 \sin x \cos x$ $2 \log \sin x - \log \sin 2x = \log \sin x - \log \cos x - \log 2$	✓ ½
2	So $I = \int_0^{\pi/2} \log \sin x \, dx - \int_0^{\pi/2} \log \cos x \, dx - \frac{\pi}{2} \log 2$	✓ ½
3	Replace x by $\frac{\pi}{2} - x$ in the first integral $\int_0^{\pi/2} \log \sin x \, dx = \int_0^{\pi/2} \log \cos x \, dx$	✓ ½
4	The first two integrals cancel $I = -\frac{\pi}{2} \log 2$	✓ ½

Answer (iii) $-\frac{\pi}{2} \log 2$

Watch out: In (iii), trying to find $\int_0^{\pi/2} \log \sin x \, dx$ separately; it is not needed, it cancels.